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Past, Present, Future, and Special Relativity

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ABSTRACT

The open future view is the common-sense view that there is an ontological difference between the past, the present, and the future in the sense that the past and the present are real, whereas the future is not yet a part of reality. In this paper we develop a theory in which the open future view is consistently combined with special relativity. Technically, the heart of our contribution is a logical conservativity result showing that, although the open future view is not definable inside the causal geometry of Minkowski space-time, it can be conservatively added to it.

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1 Introduction

In this paper the implications of the Special Theory of Relativity (STR) to philosophical views about the ontological status of the past, the present and the future will be discussed. It will be shown that the open future view, which claims that there is an ontological difference between the past, the present and the future, is *not* incompatible with the STR as has often been claimed. First, we present the open future view and the argument based on the STR against it.

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Then, developing the notion of extended space-time, we show how to restate the open future view so as to accommodate special relativity. Letting R be a binary relation between events such that eRe′ means 'taking e to be here-now, e′ is realized (that is, is in the past or in the present)', we show that R is not definable in terms of the geometric structure of Minkowski space-time, and define an extended space-time as Minkowski space-time extended by a new primitive relation R. In terms of R we define the relation of 'being in the present of' in a way that captures the intuitive notion of present described by C. D. Broad in *Scientific Thought*: 'Nothing has happened to the present by becoming past except that fresh slices of existence have been added to the total history of the world. The past is thus as real as the present. On the other hand, the essence of a present event is, not that it precedes future events, but that there is quite literally *nothing* to which it has the relation of precedence.'[1] Our main technical contribution is the logical conservativity result showing that Minkowski space-time can conservatively be extended by R.

2 Ontological separation of space-time

In this section the notion of *ontological separation of space-time*, based on the intuition that the future is ontologically different from the present and the past, is introduced.

2.1 Ontological differences between past, present, and future

According to our ordinary conception, there is an ontological asymmetry between the past, the present and the future. Present events are usually regarded as those which exist—they are occurring, they are real, whereas past events are those which have ceased to exist, and future events are those which are yet to exist. We ordinarily think that what is realized, that is, what is or has become fully real, namely the present and the past, cannot lose its reality, whereas the future, which is not yet real, can and will become so.

This ordinary conception is the basis of the *metaphysical doctrine of the open future*. According to the version of this doctrine, defended by C. D. Broad in his *Scientific Thought*,[2] reality consists of everything that has happened so far—the past is as real as the present.

The doctrine of the openness of the future should not be confused with the *doctrine of indeterminism*, which teaches that the future is 'open' in the sense that there are many alternative possible futures, since past and present do not fully fix what future events will occur. Even if there is only one possible

[1] Broad [1923], p. 66. [2] Broad [1923].

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course of history, the future events can be considered to be not yet a part of reality.

The above ontological differences between the past, the present and the future can be captured by distinguishing the present and the past as what is realized, that is what is or has become real, from the future which is not realized.

2.2 Ontological differences between past, present, and future in space-time

The status of the doctrine of the open future in the context of Minkowski space-time will be examined by discussing the notion of ontological separation of space-time. This notion we define as follows:

Minkowski space-time is ontologically separated with respect to an event e if, assuming that e is here and now, every other event e' is either:

(a) in the past or in the present—which is taken to mean that e' is realized, or

(b) in the future—which is taken to mean that e' is not realized.

Minkowski space-time is ontologically separated if it is ontologically separated with respect to every event e.

3 The argument from STR

In the opinion of many authors the STR provides a strong argument against the ontological separation of space-time.[3] The heart of the argument is the claim that the STR implies the relativization of the ontological separation of space-time to a reference frame, but the relativized ontological separation implies an unacceptable frame-dependent concept of existence. In the present section the argument is first formulated and then analysed.

3.1 The argument

To conclude that the STR implies that the notion of 'ontological separation of space-time' becomes relativized to a frame of reference, consider the following situation. Let O_1 and O_2 be two observers in uniform relative motion. Suppose O_1 and O_2 agree that event e_1 occurs 'here and now'—take e_1 to be their meeting (see Figure 1). Let e_2 be an event space-like separated from e_1—say, the birth of a star—and simultaneous with e_1 for O_1. According to the

[3] See, for example, Rietdijk [1966]; Putnam [1967]; Weingard [1972]; Mellor [1974]; Maxwell [1985].

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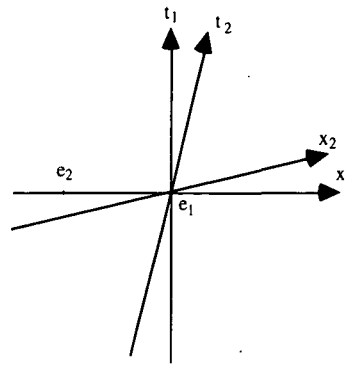


Fig 1.

STR, e_2 is not simultaneous with e_1 for O_2. Assume that O_1 and O_2 are in such relative motion that e_1 is earlier than e_2 for O_2 (see Figure 1). Then, only relative to O_1 is the star birth occurring 'now', and for the two observers space-time is *differently* ontologically separated with respect to e_1 (see Section **2.2**). In general, any change of the velocity of O_2 will result in a different separation of space-time with respect to e_1 for O_2. Since there are no preferred frames of reference, the separation of space-time depends on a frame of reference. Relative to O_1 event e_2 is realized, ontologically fixed and definite. Relative to O_2 the star birth is an event not (yet) realized. Furthermore, reality itself will be different depending on the reference frame! Relative to observer O_1 the star exists 'now', while relative to observer O_2 the star does not (yet) exist.

3.2 Analysis of the argument

The argument is based on the acceptance of the following three assumptions:

(1) the following two statements are equivalent:
  (i) With respect to e, that is, taking e to be here and now, e′ is in the present.
  (i′) e is simultaneous with e′.
(2) the following two statements are equivalent:
  (ii) With respect to e, e′ is in the past.
  (ii′) e′ is earlier than e.
(3) the following two statements are equivalent:
  (iii) With respect to e, e′ is in the future.
  (iii′) e is earlier than e′.

Since according to the STR statements (i′)–(iii′) are true only relative to a frame of reference, the acceptance of assumptions (1)–(3) implies that statements (i)–(iii) are true only relative to a frame of reference.

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A way to refute the argument from the STR is to argue that the STR is not a theory about the nature of time, but about the observable behaviour of light rays, rigid bodies and the like.[4] We, however, argue against the argument from the STR by developing what might be called the *'let's face it!'* approach. This approach assumes that the STR is a well-confirmed physical theory *about time*. In the next section the core of this assumption and one of its consequences will be briefly explained. The development of the 'let's face it!' approach will lead to the claim that the three assumptions of the argument from the STR (statements (1)–(3) in this section) do not hold. However, it will be claimed that the three assumptions do hold if e and e' are causally connectable events.

4 STR as a theory of time

That the STR is a well-confirmed theory about time basically means that the STR-temporal relations ('STR-simultaneous with', 'STR-earlier than', 'STR-later than', 'STR-absolutely earlier than', 'STR-absolutely later than', and all relations definable in terms of these relations) are truly temporal and that time has the nature attributed to it by the STR. Assuming that time described in the STR is the only time, events that are temporally related to each other are related by the STR temporal relations.

To see a very important consequence of the above statement, the relationship between temporal relations and causal relations has to be mentioned. Event e_1 is in *the STR causal relation* $<$ to event e_2 ($e_1 < e_2$), if event e_1 *can* causally influence e_2, that is, if a signal emitted at e_1 *can* reach e_2. In the standard model of the space-time geometry of the STR, namely the set $\mathbf{R}^4$ of 4-tuples $e = (x,y,z,t)$ of real numbers together with its natural vector space structure and the Minkowski metric $I^2(e_1, e_2) = (t_2 - t_1)^2 - (x_2 - x_1)^2 - (y_2 - y_1)^2 - (z_2 - z_1)^2$, '$e_1 < e_2$' is defined by the condition that the vector from e_1 to e_2 is a time-like ($I^2(e_1, e_2) > 0$) or a light-like vector ($I^2(e_1, e_2) = 0$) with $(t_2 - t_1) > 0$. The Lorentz transformation, that is, the mapping that translates the coordinate system (reference frame) of one observer into that of another, preserves the Minkowskian metric. Hence, the STR causal relation is frame-invariant. As is well known, A. Robb has shown that the space-time geometry of the STR is axiomatizable using the STR causal relation as the only primitive notion.[5] Robb's axiomatization is categorical, that is, every model of Robb's theory is isomorphic to the standard numerical model. To sum up, Robb's result shows that all STR-temporal notions are definable in terms of $<$.

The philosophical consequences of Robb's axiomatization will not be discussed here. For the purposes of this paper, Robb's result is used to conclude that any temporal relation is definable in terms of the STR causal relation.

[4] *See e.g. Smith [1993].* [5] *Robb [1936].*

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5 Ontological separation and causality

The lesson of 'the argument from the STR' is that the ontological separation of space-time has to be frame-independent. One way to ensure this is to define the ontological separation in terms of the (invariant) geometry of Minkowski space-time. Is such a definition possible? In this section, it will be concluded that the ontological separation of space-time is not definable in terms of the invariant geometrical framework.

5.1 Preliminaries

To consider how the ontological separation of space-time might be defined in terms of the STR causal relation, we need to assume that space-time is time-oriented. Space-time is time oriented if the set of time-like and light-like vectors of space-time is partitioned into the set of 'future-oriented' vectors and the set of 'past-oriented' ones. Minkowski space-time contains no such distinguished orientation—temporal orientation is not definable from the Minkowski metric. But, Minkowski space-time is time orientable. Such an orientation is determined by any arbitrary chosen coordinate system. The issues concerning the existence of time orientation will not be discussed here. So from now on, by 'Minkowski space-time' we mean 'time-oriented Minkowski space-time'.

For any event e, the notions of *'the causal past of e'*, *'the causal future of e'*, and *'the causal elsewhere of e'* are defined as follows:

(i) 'The causal past of e' is the set of all events e' such that the vector from e to e' is a time-like or a light-like vector oriented towards the past (that is, the set of all e' such that $e' < e$ holds).

(ii) 'The causal future of e' is the set of all events e' such that the vector from e to e' is a time-like or a light-like vector oriented towards the future (that is, the set of all e' such that $e < e'$ holds).

(iii) 'The causal elsewhere of e' is the set of all events e' such that the vector from e to e' is space-like (that is, the set of all e' such that neither $e' < e$ nor $e < e'$ nor $e = e'$ hold). (See Figure 2.)

To summarize: up to now, two groups of notions have been mentioned:

(1) the notions of 'causal past of e', 'causal future of e', and 'causal elsewhere of e';

(2) the notions of 'past and present with respect to e' and 'future with respect to e', where 'with respect to e' means 'taking e to be here and now'. (Recall from Section **2.2** that 'the past and the present with respect to e' is the set of all events e' such that with respect to e, e' is realized, while 'the future with respect to e' is the set of all events e' such that with respect to e, e' is not realized.)

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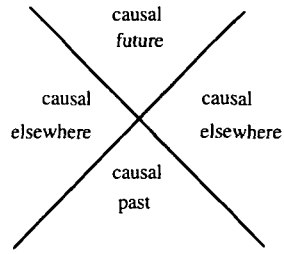


Fig 2.

Let eRe′ stand for 'with respect to e, e′ is realized' (as introduced in Section **2.2**). It will simply be assumed that the relation R is transitive, since it is generally considered to be uncontroversial.[6]

5.2 Defining ontological separation in terms of causality

Here follow some constraints that the notion of the ontological separation of space-time should satisfy.

Constraint (a) With respect to e, all events in the causal future of e are not realized.
The reasoning behind the constraint (a) is the following. If e′ is in the causal future of e, then e could be the cause of e′. Since only future events can be affected, e′ is in the future of e. Hence, e′ is not realized with respect to e.

Constraint (b) With respect to e, e itself is realized.
The argument for constraint (b) is the following. We have already assumed that e is here and now. So, e belongs to the present of e. Hence, e is realized with respect to e.

Constraint (c) With respect to e, all events from the causal past of e are realized.
The reasoning behind the constraint (c) is the following. If e′ is in the causal past of e, then e′ could be the cause of e. Since only the past can influence the present, e′ belongs to the past of e. Hence, e′ is realized with respect to e.

Constraints (a) and (c) reinforce an earlier remark that in order to see how the ontological separation of space-time might be defined in terms of the STR causal relation, we need to assume that space-time is time-oriented. Without an objectively preferred time-orientation, constraints (a) and (c) are contradictory. Hence, R is certainly not definable in Minkowski space-time without a time direction.

[6] See Putnam [1967]; Weingard [1972]; Christensen [1981]; Maxwell [1985]; Stein [1991].

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Now, what about the events that are neither in the causal future of event e nor in the causal past of e. That is, what about the events that are in the causal elsewhere of e? Are these events realized with respect to e or not realized with respect to e? Do all these events have the same ontological status with respect to e, or do different events from the causal elsewhere of e have different ontological status with respect to e? The fact is that if the notion of ontological separation that satisfies the above constraints (a)–(c) is definable in terms of the STR causality relation, then with respect to any e, all events from the causal elsewhere of e are not realized with respect to e. This fact follows from the following theorem proved by H. Stein:

> If $\mathcal{R}$ is a reflexive and transitive relation on a Minkowski space (of any number of dimensions—of course at least two), invariant under automorphisms that preserve the time-orientation, and if $\mathcal{R}$ab holds for some pair of points (a,b) such that ab is a past-pointing (timelike or null) nonzero vector, then for any pair of points (x,y), $\mathcal{R}$xy holds if and only if xy is a past-pointing vector.[7]

If a relation is definable in terms of the geometric structure of Minkowski space-time, then R must be invariant with respect to all automorphisms of that structure. Now, taking the relation R to be $\mathcal{R}$, the following version of Stein's theorem holds in our framework:

> If the relation R is definable in terms of the geometric structure of Minkowski space-time, then for any two events e and e′ it holds that: eRe′ iff either e = e′ or e′ is in the causal past of e.

Hence, if the relation R is definable, then the notion of ontological separation of space-time due to R satisfies the following constraint.

Constraint (d) With respect to e, all events from the causal elsewhere of e are not realized.

Argument against constraint (d)

The absence of any possible causal influence between two events e and e′ cannot be (either the reason for e′ to be realized with respect to e, or) the reason for e′ to be not realized with respect to e.

Imagine the following situation. The occurrence or the nonoccurrence of a seafight depended solely on the 'free' decision of an admiral. The admiral actually decided—event e_2 in Figure 3—that there was going to be the seafight—event e_3 in Figure 3. The event e_2 is very 'far' from event e_1, but in its causal past. The seafight is in the causal future of the admiral's decision, but the seafight is in the elsewhere of e_1. At e_1 an observer knows about the admiral's decision. It is hard to see why it should be impossible that e_3 has

[7] Stein [1991], pp. 149–50. Another proof of this theorem, different from Stein's proof, can be found in Clifton and Hogarth [1995]. However, Clifton and Hogarth weaken the last premise of Stein's theorem by assuming that the vector ab is a time-like past-pointing vector.

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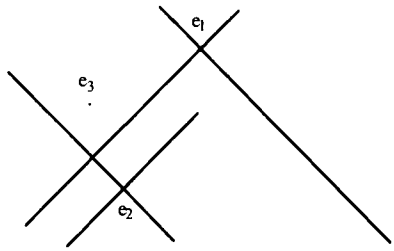


Fig 3.

already become, i.e. is realized with respect to e_1, although the observer still has to wait for a while to receive the message saying that the seafight did occur. The fact that at e_1 the message has not yet been received does not imply that the seafight is not realized. Furthermore, it seems that, assuming that event e_2 is distant enough from the observer at e_1 —so that when the observer receives the message about the admiral's decision he cannot intuitively think that the admiral is still alive—there will be a good intuitive reason to say that e_3 is realized with respect to e_1.

Hence, constraint (d) is unacceptable.[8] Consider for the moment the constraint opposite to constraint (d), namely:

Constraint (e): With respect to e, all events from the causal elsewhere of e are realized.

This constraint contradicts constraint (a).

Argument against constraint (e)

Consider events e, e′ and e″ such that e is in the causal future of e, and e″ is in the elsewhere of both e and e′ (see Figure 4). Then, e and e′ are in the causal elsewhere of e″. Now, constraint (e) implies the following: with respect

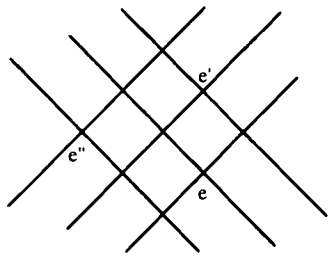


Fig 4.

[8] For a further discussion of constraint (d), see Maxwell [1985, 1993] and Stein [1991]. Stein accepts constraint (d), while Maxwell argues against it.

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to e, e'' is realized; with respect to e'', e' is realized. By the transitivity of the relation R, it follows that with respect to e, e' is realized. This contradicts constraint (a).

5.3 Conclusion

The argument against constraint (d) shows that for every event e, some events in the causal elsewhere of e might be realized with respect to e. The argument against constraint (e) shows that some events in the causal elsewhere of e might be not realized with respect to e. To sum up, different events in the causal elsewhere of e can have different ontological status with respect to e. This view on the 'elsewhere' events implies that we do need an R relation in order to discriminate between realized and not realized 'elsewhere' events.

Since constraint (d) is shown to be unacceptable, Stein's theorem implies that the relation R is not definable in terms of the geometry of Minkowski space-time. Since $<$ is definable in terms of Minkowski metric, it follows that the relation R is not definable in terms of the causality relation, that is, that *the ontological separation of Minkowski space-time is not definable in terms* of $<$.

6 Extended space-time

In this section the relation R will be introduced in the structure of space-time as a new primitive relation on events. For this purpose, oriented Minkowski space-time will be represented as a pair $\langle W, \leq \rangle$ where W is the set of non-empty set W of (point) events subject to the STR-causal relation $\leq$ which is reflexive (Axiom 1(i) below), transitive (Axiom 1(ii)), directed (Axiom 1(iii)–(iv)), antisymmetric (Axiom 1(v)) and strictly dense (Axiom 1(vi)). The relations of 'being in the elsewhere of' and 'being in the causal future of' will be defined in terms of the relation $\leq$, and the following three geometrical facts about Minkowski space-time will be added as Axioms:

(1) *the elsewhere of any event is not empty* (Axiom 1(vii));
(2) for any two events e_1 and e_2, if e_2 is in the elsewhere of e_1, there is a third event e_3 which is in the causal future of e_1 and in the elsewhere of e_2 (Axiom 1(viii));
(3) for any two events e_1 and e_2, if e_2 is in the elsewhere of e_1, there is a third event e_3 which is in the causal past of e_1 and in the elsewhere of e_2 (Axiom 1(ix)).

After that, a number of new Axioms on the relations $\leq$ and R will be stated, leading to the desired separation of space-time.

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6.1 Extended space-time frame

An *extended space-time frame* is an ordered triple $\langle W, \leq, R\rangle$ in which W is the set of point events, and $\leq$ and R are binary relations on W. Before stating the axioms on the STR-causal relation, we define 'e_2 is in the causal future of e_1' and 'e_2 is in the elsewhere of e_1' in terms of the relation $\leq$ in the following way:

Definition 1. $e_1 < e_2$ iff $(e_1 \leq e_2 \wedge \neg(e_1 = e_2))$.

Definition 2. e_1 ELS e_2 iff $(\neg(e_1 \leq e_2) \wedge \neg(e_2 \leq e_1))$.

Axiom 1.
(i) $\leq$ is reflexive.
(ii) $\leq$ is transitive.
(iii) $\leq$ is upward directed (any e_1, e_2 have a common upper bound e_3: $e_1 \leq e_3$ and $e_2 \leq e_3$).
(iv) $\leq$ is downward directed (any e_1, e_2 have a common lower bound e_3: $e_3 \leq e_1$ and $e_3 \leq e_2$).
(v) $\leq$ is antisymmetric (whenever $e_1 \leq e_2$ and $e_2 \leq e_1$ then $e_1 = e_2$).
(vi) $\leq$ is strictly dense (whenever $e_1 < e_2$ there exist e_3 such that $e_1 < e_3$ and $e_3 < e_2$).
(vii) the elsewhere of any event is not empty.
(viii) for any two events e_1 and e_2, if e_1 ELS e_2, then there is a third event e_3 such that $e_1 < e_3$ and e_2 ELS e_3.
(ix) for any two events e_1 and e_2, if e_1 ELS e_2, then there is a third event e_3 such that $e_3 < e_1$ and e_2 ELS e_3.

A full Robb-style axiomatization of Minkowski space-time geometry could have been taken as Axiom 1, but there is no need for that. Axiom 1 in the given form is a description of Minkowski space-time sufficient for our purpose. Here follow some ready-proved theorems on the connections between the relations $\leq$, $<$ and ELS.

Theorem 1.
(i) ELS is irreflexive and symmetric.
(ii) $\forall e_1,e_2(e_1 \leq e_2 \vee e_2 \leq e_1 \vee e_1 \text{ ELS } e_2)$.
(iii) $\forall e_1,e_2(e_1 \text{ ELS } e_2 \rightarrow \exists e_3(e_1 < e_3 \wedge e_2 < e_3))$.
(iii′) $\forall e_1,e_2(e_1 \text{ ELS } e_2 \rightarrow \exists e_3(e_3 < e_1 \wedge e_3 < e_2))$.
(iv) $\forall e_1 \exists e_2(e_1 < e_2)$.
(v) $\forall e_1 \exists e_2(e_2 < e_1)$.

We proceed to state the axioms connecting the relation $\leq$ and the new primitive relation R. The next two axioms reflect the constraints (a)–(c) in Section **5.2**.

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Axiom 2. $\forall e_1,e_2(e_1 \leq e_2 \rightarrow e_2\ R\ e_1)$.
Axiom 2 says that if e_1 is in the causal past of e_2 or $e_1 = e_2$, then with respect to e_2, e_1 is realized.

Theorem 2.
(i) $\forall e(eRe)$.
(ii) $\forall e_1,e_2,e_3(e_1 \leq e_2 \wedge e_2 \leq e_3 \rightarrow e_3\ R\ e_1)$.
(iii) $\forall e_1,e_2(e_1\ R\ e_2 \rightarrow (e_2 \leq e_1 \vee e_1\ ELS\ e_2)$.

Axiom 3. $\forall e_1,e_2(e_1 < e_2 \rightarrow \neg(e_1\ R\ e_2))$.
Axiom 3 says that if e_2 is in the causal future of e_1, then with respect to e_1, e_2 is not realized.

Theorem 3.
(i) $\forall e_1,e_2(\neg(e_1\ R\ e_2) \rightarrow (e_1 < e_2 \vee e_1\ ELS\ e_2)$.
(ii) $\forall e_1,e_2,e_3(e_1 < e_2 \wedge e_2 \leq e_3 \rightarrow \neg(e_1\ R\ e_3))$.

Transitivity of the relation R has been accepted in Section **5.1**. It is taken here as an axiom on R.

Axiom 4. $\forall e_1,e_2,e_3(e_1\ R\ e_2 \wedge e_2\ R\ e_3 \rightarrow e_1\ R\ e_3)$.

Theorem 4.

(i) $\forall e_1,e_2,e_3(e_1\ R\ e_2 \wedge e_3 \leq e_2 \rightarrow e_1\ R\ e_3)$.

Theorem 4(i) says that if e_2 is realized with respect to e_1, then everything that is in the causal past of e_2 is realized with respect to e_1.

(ii) $\forall e_1,e_2,e_3(\neg(e_1\ R\ e_2) \wedge e_2 \leq e_3 \rightarrow \neg(e_1\ R\ e_3))$.

Theorem 4(ii) says that if e_2 is not realized with respect to e_1, then everything that is in the causal future of e_2 is not realized with respect to e_1. Before taking a look at some further consequences of the acceptance of Axiom 4, R_e is defined to be the set of all events that are realized with respect to event e.

Definition 3. $R_e = \{e' | e\ R\ e'\}$.

(iii) $\forall e,e'(e\ R\ e' \times R_{e'} \amalg R_e)$.

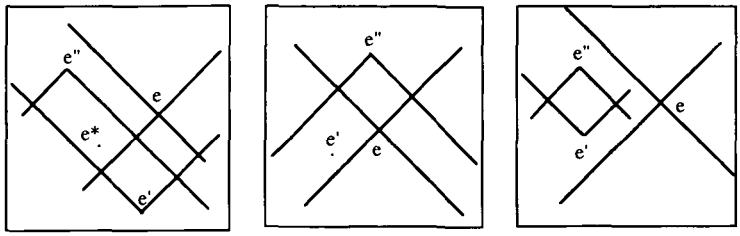
(iv) $\forall e,e'(R_e = R_e \times e\ R\ e' \wedge e'R\ e)$.

Finally, we state the last Axiom.

Axiom 5.
$\forall e, e', e''(e' < e'' \wedge e\ R\ e' \wedge \neg(e\ R\ e'') \rightarrow \exists e^*(e' \leq e^* \wedge e^* < e'' \wedge e\ R\ e^* \wedge e^*\ R\ e))$.

Let us first see what Axiom 5 says about a situation in which events e, e' and e'', such that $e' < e$ and $eELSe''$ (see Figure 5(a)), satisfy the antecedent of Axiom 5. In such a situation Axiom 1(vi) implies that there is an event which is

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(a) (b) (c)

Fig 5.

causally between e′ and e″. Now, Axiom 5 says that either eRe′ and e′Re hold, or eRe* and e*Re hold for some e* which is causally between e′ and e″. Since in the situation in Figure 5(a) event e′ is in the causal past of e, Axiom 3 implies ¬(e′Re). Moreover, event e is not realized with respect to any event which is causally between e′ and e″, and which is in the causal past of e. Thus, it follows that eRe* and e*Re hold for some e* which is in the elsewhere of e, and which is causally between e′ and e″. Now, taking into account Theorem 4(iv), Axiom 5 implies that $R_e = R_{e*}$ for some such e*. According to our definition of the ontological separation of Minkowski space-time, this says that with respect to some event which is causally between e′ and e″, and in the elsewhere of e, space-time is ontologically separated in the same way as it is with respect to e.

To see what are the other situations in which Axiom 5 is applicable, note that the condition eRe′ in its antecedent implies, by Theorem 2(iii), that either $e' \leq e$ or eELSe′. Also, the condition ¬(eRe″) implies, by Theorem 3(i), that either $e < e''$ or eELSe″. Now, it is easy to see that when events e, e′ and e″ are such that $e' \leq e$, $e < e''$ and the antecedents of Axiom 5 holds, the consequent of Axiom 5 can be proved using Theorem 2(i). But, for the sake of elegance this case is not excluded from the axiom. Thus, Axiom 5 says something new only for two more cases: when e′ELSe and $e < e''$ (see Figure 5(b)), and when e′ELSe and eELSe″ (see Figure 5(c)).

To sum up, Axiom 5 says that between any two causally connectable events such that the first one is realized with respect to e, whereas the second one is not realized with respect to e, there is an event e* such that space-time is ontologically separated in the same way with respect to e and e*. This axiom is natural and, as we shall see, rich in consequences. The proofs of Theorem 5, Theorem 6(ii)–(vi), Theorem 7(i) and (ii), Theorem 8(ii), (iv), (vi) and (vii) below depend essentially on Axiom 5. The proofs of most of the theorems that follow are given in the Appendix.

Theorem 5.

(i) $\forall e \exists e' (e \text{ ELS } e' \wedge e \text{ R } e')$.

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Theorem 5(i) is inconsistent with the constraint (d) in section 5.2. Theorem 5(i) says that it is not the case that all events from the elsewhere of e are unrealized with respect to e.

(ii) $\forall e \exists e(e \text{ ELS } e \wedge \neg(e \text{ R } e'))$.

Theorem 5(ii) is inconsistent with the constraint (e) in Section **5.2**. Theorem 5(ii) says that it is not the case that all events from the elsewhere of e are realized with respect to e. This theorem, together with Theorem 5(i), implies that with respect to every event e, some elsewhere-events are realized and some elsewhere-events are not realized.

(iii) $\forall e,e'(e \text{ R } e' \vee e' \text{ R } e)$.

At this point we want to introduce the relation PRES. 'e PRES e′' will stand for 'with respect to e, e′ is present'. We have already seen that two events e and e′ are realized from each other's perspective if and only if space-time is ontologically separated in the same way with respect to e and with respect to e′. Hence, it is intuitively plausible to assume that two events that are realized from each other's perspective belong to the same present.

Definition 4. e PRES e′ iff (e R e′ ∧ e′R e).

Theorem 6.

(i) $\forall e,e'(e \text{ PRES } e' \leftrightarrow R_e = R_{e'})$.
(ii) $\forall e,\ e',\ e''(e' < e'' \wedge e \text{ R } e' \wedge \neg(e \text{ R } e'') \rightarrow \exists e^*(e' \leq e^* \wedge e^* < e'' \wedge e \text{ PRES } e^*))$.

Theorem 6(i) is a consequence of Definition 4 and Theorem 4(iv). Theorem 6(ii) follows from Axiom 5 simply by definition of PRES. This theorem ensures that the intuitive notions of past, present and future, as described by Broad, can be defined in terms of R in such a way that they fit well with the notions of causal past, causal future and causal elsewhere. According to our ordinary conception, the present is the boundary between the past and the future, and hence between any past and any future there always exists a present. In the relativistic context, where the notion of causal connectability plays an important role, that ordinary conception is reflected in Theorem 6(ii). Theorems 6(iii)–(vi) below are consequences of Axiom 5 (we omit their proofs).

(iii) $\forall e \forall e'((e \text{ ELS } e' \wedge \neg(e \text{ R } e')) \rightarrow \exists e''(e \text{ ELS } e'' \wedge e'' < e' \wedge e \text{ PRES } e''))$.
(iv) $\forall e \forall e'(\neg(e \text{ R } e')) \rightarrow \exists e''(e'' < e' \wedge e \text{ PRES } e''))$.

Theorem 6(iv) claims that if e′ is not realized with respect to e, then a present-event with respect to e exists in the causal past of e′.

(v) $\forall e \forall e'((e \text{ ELS } e \wedge e \text{ R } e') \rightarrow \exists e''(e \text{ ELS } e'' \wedge e' \leq e'' \wedge e \text{ PRES } e''))$.
(vi) $\forall e, e'(e \text{ R } e' \rightarrow \exists e''(e' \leq e'' \wedge e \text{ PRES } e''))$.

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Theorem 6(vi) claims that if e_1 is realized with respect to e, then either e_1 is a present-event with respect to e or a present-event with respect to e exists in the causal future of e_1.

The next theorem states how the relation PRES and Broad's conception of the present are related.

Theorem 7.
(i) $\forall e,e'(e \text{ PRES } e' \rightarrow (e' = e \text{ or } e \text{ ELS } e'))$.
(ii) $\forall e(e \text{ PRES } e)$.
(iii) $\forall e,e'(e \text{ PRES } e' \rightarrow e' \text{ PRES } e)$.
(iv) $\forall e,e',e''(e \text{ PRES } e' \wedge e' \text{ PRES } e'' \rightarrow e \text{ PRES } e'')$.
(v) $\forall e,e'((e \text{ PRES } e') \rightarrow (e \text{ R } e' \wedge \forall e''(e' < e'' \rightarrow \neg(e \text{ R } e''))))$.

The proofs of Theorems 7(i) and (v) involve Axiom 5. Theorem 7(ii) follows from Theorem 2(i), while Theorem 7(iv) is a consequence of Axiom 4. Theorem 7 shows that the relation PRES formalizes Broad's conception of present in space-time framework. Theorems 7(ii)–(iv) say that PRES is a reflexive, symmetric and transitive relation. Thus, PRES is an equivalence relation. This captures Broad's description of the present as a 'slice of existence'. The PRES-equivalence classes correspond to Broad's 'slices of existence'. Besides, Theorem 7(v) shows that relation PRES captures what Broad would call 'the essence of a present event' with respect to an event e: if e' is present with respect to e, then e' is realized with respect to e, and nothing to which e' has the $<$ relation is realized with respect to e.[9]

Since the relation PRES is an equivalence relation, the set of events W is partitioned by PRES into equivalence classes.

Definition 5. $[e] = \{e' | \text{ e PRES } e'\}$.

We can now show that the set of PRES-equivalence classes is linearly ordered by the relation $\propto$, defined in the following way.

Definition 6. $[e_1] \propto [e_2]$ iff $\neg(e_1 \text{ R } e_2)$.

Theorem 8.
(i) If $e' \in [e_1]$, $e'' \in [e_2]$ and $[e_1] \propto [e_2]$, then $\neg(e' \text{ R } e'')$.
(ii) For every e, set R_e is the union U of [e] and the causal pasts of all members of [e].
(iii) $\propto$ is irreflexive.
(iv) $\propto$ is transitive.

[9] Let us mention that our approach could be based on the extension of Minkowski space-time by the primitive equivalence relation PRES. Then, the relation R is definable as follows: eRe' iff $\exists e^*(e \text{ PRES } e^* \wedge e' \leq e^*)$. The undefinability of R in terms of the geometric structure of Minkowski space-time then follows from the theorem of van Benthem saying that the only equivalence relations definable in Minkowski space-time are the universal and the identity relation (van Benthem [1983], p. 25).

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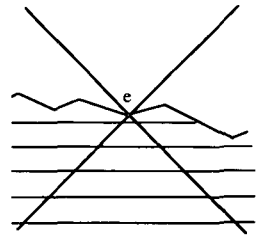


Fig 6.

(v) For every two equivalence classes $[e_1]$ and $[e_2]$, either $[e_1] \propto [e_2]$ or $[e_2] \propto [e_1]$ or $[e_1] = [e_2]$ holds.
(vi) $\propto$ is dense.
(vii) $[e_1] \propto [e_2]$ iff $R_{e1} \subset R_{e2}$.

Proofs of Theorems 8(ii), (iv), (vi) and (vii) depend on Axiom 5, while proofs of Theorems 8(i), (iii), and (v) depend on Axiom 4, Theorem 2(i) and Axiom 3 respectively. Theorem 8 makes it now easy to see that the relation $\angle$ on the set of R_e-sets defined by the condition '$R_{e1} \angle R_{e2}$ iff $R_{e1} \subset R_{e2}$' is linear.

For an example of extended space-time see Figure 6.

6.2 Conservativity with respect to extensions of Minkowski space-time

It has already been shown that the relation R is not definable in terms of the STR causal relation. We show now the conservativity of the relation R over Robb's theory. To prove this we use the fact that Robb's theory has a model consisting of the set $\mathbf{R}^4$ of 4-tuples of real numbers together with the Minkowski metric (see Section **4**). We formulate our result wider so that the result applies to subtheories of Robb's theory which allow adding a sufficiently large part of the theory of real numbers with the Minkowski metric. We first establish Proposition 1. The conservativity result will be a corollary of Proposition 1.

To explain the claim of Proposition 1, let us first describe theories **T** and $\mathbf{T}^{\mathbf{Real}}$. Theory **T** is a subtheory of Robb's theory. Theory **T** is extended into theory $\mathbf{T}^{\mathbf{Real}}$ by four projection functions which attach a unique four-tuple of real numbers to each event. Condition (i) in Proposition 1 below ensures that the same four-tuple cannot be attached to different events. In this way theory **T** is *syntactically* extended with a particular frame of reference—each event becomes supplied with 'coordinates' of the frame of reference given by real numbers. Proposition 1 claims that in theory $\mathbf{T}^{\mathbf{Real}}$ our realization relation R can be defined. An obvious definition of R is to let eRe' hold if and only if e' is

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simultaneous with or earlier than e in the frame of reference contained in the theory. However, many other definitions are available. Consider again the two-dimensional example in Figure 6 where the set R_e of all events realized with respect to e is presented. In fact, any set of events would do as a 'good' R_e, provided only that the upper bound of the set belongs to the 'elsewhere' of e in the frame of reference contained in the theory. In accordance with that, the proof of Proposition 1 does not simply give a particular 'good' definition of R in $\mathbf{T}^{\mathbf{Real}}$, that is, a definition of R such that R satisfies Axioms 2–5. The proof shows that a whole class of definitions, satisfying the above mentioned condition that the upper bound of each R_e set belongs to the 'elsewhere' of e, is 'good'.

Proposition 1.
Let T be a subtheory of Robb's theory, and let T^{Real} *be the extension of T by the first order theory of* $(\boldsymbol{R}, 0, +, -, \cdot, \leqslant)$, *and by projection functions* $\pi_1,\ldots,\pi_4$ *from events to real numbers such that:*

(i) $(\pi_1(e)=\pi_1(e')\wedge\pi_2(e)=\pi_2(e')\wedge\pi_3(e)=\pi_3(e')\wedge\pi_4(e)=\pi_4(e'))\rightarrow e=e'$,
(ii) $e<e'\rightarrow\pi_4(e)\lessdot\pi_4(e')$,
(iii) $\forall k,l,m,n\exists e(\pi_1(e)=k\wedge\pi_2(e)=l\wedge\pi_3(e)=m\wedge\pi_4(e)=n)$,
(iv) $\forall e,e'\forall n((e<e'\wedge\pi_4(e)\lessdot n\wedge n\lessdot\pi_4(e')\rightarrow\exists e^*(e<e^*\wedge e^*<e'\wedge\pi_4(e^*)=n))$,

with e,e′,e ranging over events, and k,l,m,n ranging over reals.*
Then, a relation R, satisfying Axioms 2–5, is definable in $\mathbf{T}^{\mathbf{Real}}$.

Proof. We approach the definition of R via function F from pairs of events to events, which, intuitively speaking, does the following. Consider event e. Function F takes a pair (e, e′) to the event which: (a) occurs at the 'same place' as e′ in the frame of reference contained in $\mathbf{T}^{\mathbf{Real}}$, and (b) is in the ontological present of e. (It is easy to visualize this in a two-dimensional frame whose root is in e.) Thus, the space coordinates, that is, the projection functions π_1,π_2,π_3 of events e′ and F(e,e′) will be the same—see condition C1 below. If function F is such that all events in the ontological present of any e belong to the 'elsewhere' of e in the frame of reference contained in $\mathbf{T}^{\mathbf{Real}}$, which will be ensured by conditions C2–C6 below, then the choice for the definition of R is obvious. We let eRe′ hold if e′ is in the ontological present or in the ontological past of e. To determine whether that is the case, we need to compare the time coordinate π_4(e′) of e′ with the time coordinate π_4(F(e,e′)) of event F(e,e′). Recall that F(e,e′) is in the ontological present of e, and that e′ occurs at the same place as F(e,e′). Clearly, we let eRe′ hold if e′ occurs earlier than or simultaneous with F(e,e′), that is, if π_4(e′) is smaller than or equal to π_4(F(e,e′)).

Thus, take any function F from pairs of events to events definable in $\mathbf{T}^{\mathbf{Real}}$

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which satisfies the following conditions.

C1. $\pi_1(F(e,e')) = \pi_1(e') \wedge \pi_2(F(e,e')) = \pi_2(e')\ \pi_3(F(e,e')) = \pi_3(e')$.
C2. $e' \leq e \rightarrow (\pi_4(e') \leq \pi_4(F(e,e')))$.
C3. $e<e' \rightarrow (\pi_4(F(e,e')) < \pi_4(e'))$.
C4. $(\pi_4(e') \leq \pi_4(F(e,e'))) \wedge (\pi_4(e'') \leq \pi_4(F(e',e''))) \rightarrow (\pi_4(e'') \leq \pi_4(F(e,e'')))$.
C5. $(F(e,e') = e') \rightarrow (F(e',e) = e)$.
C6. $e' < e'' \wedge \pi_4(e') \leq \pi_4(F(e,e')) \wedge \neg(\pi_4(e'') \leq \pi_4(F(e,e'')))$
$\rightarrow \exists e^*(e' \leq e^* \wedge e^* < e'' \wedge F(e,e^*) = e^*)$.

Condition (iii) in Proposition 1 ensures that there are many such F. A trivial example is **F** defined as follows:

$\mathbf{F}(e,e') = e^*$ where $\pi_1(e^*) = \pi_1(e'),\ldots, \pi_3(e^*) = \pi_3(e')$ and $\pi_4(e^*) = \pi_4(e)$.

This **F** satisfies condition C1 by definition. That **F** satisfies C2 follows from the transitivity of the real numbers ordering $\leq$. Condition (ii) of Proposition 1 implies that **F** satisfies conditions C3 and C4, whereas the definition of **F** implies that **F** satisfies condition C5. Finally, to see that **F** satisfies condition C6, assume $e' < e''$, $\pi_4(e') < \pi_4(\mathbf{F}(e,e'))$ and $\neg(\pi_4(e'') \leq \pi_4(\mathbf{F}(e,e'')))$. By the definition of **F** and properties of $\leq$, it follows that $\pi_4(e') \leq \pi_4(e)$ and $\pi_4(e) < \pi_4(e'')$. If $\pi_4(e') = \pi_4(e)$, then $\mathbf{F}(e,e') = e'$, and hence C6 holds. If $\pi_4(e') < \pi_4(e)$, then condition (iv) of Proposition 1 implies $\exists e^*(e' < e^* \wedge < e^* < e'' \wedge \pi_4(e^*) = \pi_4(e))$. Then, $\mathbf{F}(e,e^*) = e^*$. Hence, C6 is satisfied.

Let R be a relation defined as follows:

$$eRe' \text{ iff } (\pi_4(e') \leq \pi_4(F(e,e'))).$$

It is easy to see that the example of function **F** given above makes eRe' hold if and only if e' is (metrically) in the present or in the past of e in the frame of reference contained in the theory.

That in the general case R satisfies Axioms 2–4 follows immediately from conditions C2, C3 and C4 respectively. To show that R satisfies Axiom 5, assume $e' < e''$, $(\pi_4(e') \leq \pi_4(F(e,e')))$ and $\neg(\pi_4(e'') \leq \pi_4(F(e,e'')))$. We need to show that there is e^* such that $e \leq e^*$, $e^* < e''$, eRe^* and e^*Re. By C6, there is e^* such that $e' \leq e^*$, $e^* < e''$ and $F(e,e^*) = e^*$. By C5, it follows that $F(e^*,e) = e$. Now, $F(e,e^*) = e^*$ and $F(e^*,e) = e$ imply $\pi_4(e^*) = \pi_4(F(e,e^*))$ and $\pi_4(e) = \pi_4(F(e^*,e))$ respectively. By the definition of R, eRe^* and e^*Re. ♥

A corollary of Proposition 1 is the following general conservativity result. Consider any extension $\mathbf{T}^+$ of a Robb-subtheory **T**, in any language, such that the extension $\mathbf{T}^{+,\mathbf{Real}}$ of $\mathbf{T}^+$, by the theory of real numbers as in Proposition 1, is conservative over $\mathbf{T}^+$. Then, addition of Axioms 2–5 to $\mathbf{T}^+$ is conservative over $\mathbf{T}^+$. The conservativity of Axioms 2–5 over Robb's theory is a special case of this corollary. Robb's theory is an extension of theory **T**. Also, the extension of Robb's theory by the theory of real numbers is a conservative extension, since Robb's theory holds in Minkowski space-time. Moreover, the

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general conservativity result ensures that even when Robb's theory is further extended by, say, some physical theory, addition of Axioms 2–5 stays conservative, assuming that the physical theory can be conservatively extended by the theory of real numbers.

Corollary 1.
Let $\boldsymbol{T}$ *be a a subtheory of Robb's theory, and let* $\boldsymbol{T}^{+}$ *be an extension of* $\boldsymbol{T}$*. Let* $\boldsymbol{T}^{+}$ *be extended into* $\boldsymbol{T}^{+,Real}$ *in the way in which theory T was extended to* $\boldsymbol{T}^{Real}$ *in Proposition 1: by the first order theory of* $(\boldsymbol{R}, 0, +, -, \cdot, \leq)$*, and by projection functions* $\pi_1,...,\pi_4$ *satisfying conditions (i)–(iv) as stated in Proposition 1. Then:*

> *If* $\boldsymbol{T}^{+,Real}$ *is a conservative extension of* $\boldsymbol{T}^{+}$*, then adding Axioms 2–5 to* $\boldsymbol{T}^{+}$ *gives a conservative extension of* $\boldsymbol{T}^{+}$*.*

Proof. Assume that $\mathbf{T}^{+,\mathbf{Real}}$ is a conservative extension of $\mathbf{T}^{+}$. Let $\mathbf{T}^{+,\mathrm{R}}$ be the extension of $\mathbf{T}^{+}$ by Axioms 2–5. We show now that $\mathbf{T}^{+,\mathrm{R}}$ is a conservative extension of $\mathbf{T}^{+}$. Let φ be a sentence in the language of $\mathbf{T}^{+}$, and derivable in $\mathbf{T}^{+,\mathrm{R}}$. We need to show that φ is derivable in $\mathbf{T}^{+}$. Since φ is derivable in $\mathbf{T}^{+,\mathrm{R}}$, φ is derivable in $\mathbf{T}^{+,\mathrm{R},\mathbf{Real}}$. By $\mathbf{T}^{+,\mathrm{R},\mathbf{Real}}$ we mean the extension of $\mathbf{T}^{+,\mathrm{R}}$ by the first order theory of $(\mathbf{R}, 0, +, -, \cdot, \leq)$ and by projection functions $\pi_1,...,\pi_4$ as stated in Proposition 1. If the derivation of φ in $\mathbf{T}^{+,\mathrm{R},\mathbf{Real}}$ does contain occurrences of the predicate latter R, then these can be replaced by the relation R defined in $\mathbf{T}^{+,\mathbf{Real}}$ according to Proposition 1. Hence, φ is derivable in $\mathbf{T}^{+,\mathbf{Real}}$. Since we have assumed that $\mathbf{T}^{+,\mathbf{Real}}$ is a conservative extension of $\mathbf{T}^{+}$, it follows that φ is derivable in $\mathbf{T}^{+}$.♥

7 The argument from STR revisited

Our answer to the argument from the STR against the ontological separation of space-time is now clear: the three basic assumptions of the argument are not valid when the notions of past, present and future, that are introduced in this paper, are involved. On one side, these notions of past, present and future are based on the ontological distinction between realized and unrealized events. In the context of Minkowski space-time, the ontological relation R on events yields this distinction. Since R is not definable in terms of the STR causal relation, R is not a temporal but an ontological relation, and the notions of past, present and future are not temporal but ontological notions. On the other side, the frame dependent temporal notions, such as the notion of 'simultaneity', do not have any immediate implications for the ontological distinction between realized and not realized events. The frame dependent temporal notions reflect the fact that the STR causal relation is only partly informative on ontological issues: only for the events causally connectable to an event e it is certain whether they are realized with respect to the event e. Hence, the frame

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dependent temporal relations between causally nonconnectable events can differ from frame to frame without causing any 'ontological trouble'. This separation of time from ontology is the most important implication of the accommodation, presented in this paper, of the doctrine of the open future to the STR.

One possible criticism of our approach is that the new primitive relation R on the events of Minkowski space-time does not have a place in our physical theory. To answer this criticism I call on Einstein himself for help. Einstein, according to the writing of R. Carnap in his *Autobiography*, 'once ... said that the experience of the Now means something special for man, something essentially different from the past and the future, but that this important difference does not and cannot occur within physics.'[10] Although Einstein's theory forces difficult revisions of some common temporal notions, such as the notion of 'simultaneity', 'being earlier than' and 'being later than', it does not force us to give up the open future view, that is, the ontological distinction between the past, the present and the future. Still, the open future view must be reconciled with the results of physical science.

Let us conclude by mentioning that the results presented in this paper could be considered from a different basic assumption, namely from the assumption that the STR is not a theory about time. From this point of view, our results could be seen as saying that the *truly temporal* relation R is not definable in terms of the nontemporal STR notions; and thus that the nontemporal STR can be and needs to be extended by the relation R, a primitive truly temporal relation on events which introduces absolute time in the nontemporal STR. This would imply that the notions of past, present and future, defined in terms of the relation R, are the usual common-sense notions of past, present and future, and that time and ontology go together as usual. Claiming that the STR is a nontemporal theory does save common sense from trouble, but 'being in accordance with common sense' is usually not a very good criterion for interpreting scientific theories. Once we accept the STR for what, among other things, it claims to be—a theory about time—some of our common sense notions have to adapt to it.

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[10] Cited in Schlesinger [1994].

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this paper with indeterminism.) We would like to thank Frank Veltman, George Berger, an anonymous referee of this journal, and especially Dick de Jongh for helpful discussions and comments.

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Appendix

Theorem 5.

(i) $\forall e \exists e'(e \text{ ELS } e' \wedge e \text{ R } e')$.

Proof: Let e be arbitrary. By Axiom 1(vii), there is e* such that eELSe*. If eRe*, we are done. Assume $\neg$(eRe*). By Theorem 1(iii′), there is e″ such that

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$e'' < e$ and $e'' < e^*$. $e'' < e$ implies, by Axiom 2, eRe''. By Axiom 5 there is e' such that $e'' \leq e'$, $e' < e^*$, eRe' and $e'Re$. eRe' implies, by Theorem 2(iii), that either $eELSe'$ or $e' \leq e$. Since $e' < e^*$ and $eELSe^*$, $e \neq e'$. Since $e'Re$, Axiom 3 implies $\neg(e' < e)$. Hence, there is e' such that $eELSe'$ and eRe'. ♥

(ii) $\forall e \exists e'(e \text{ ELS } e' \wedge \neg(e \text{ R } e'))$.

Proof: Let e be an arbitrary event. Axiom 1(vii) implies that there is event e'' such that $eELSe''$ holds. If $\neg(eRe'')$, then we are done. So, suppose eRe''. By Theorem 1(iii), there is $e^{\#}$ such that $e'' < e^{\#}$ and $e < e^{\#}$. By Axiom 3, $\neg(eRe^{\#})$. By Axiom 5, there is e^* such that $e'' \leq e^*$, $e^* < e^{\#}$, eRe^* and e^*Re. eRe^* implies, by Theorem 2(iii), that either $e^* \leq e$ or $eELSe^*$. Since $e'' \leq e^*$ and $eELSe^*$, $e \neq e''$. Since e^*Re, Axiom 3 implies $\neg(e^* < e)$. Hence, $eELSe^*$. By Theorem 1(i), $eELSe^*$ implies e^*ELSe. Axiom 1(viii) implies that there is e' such that $e^* < e'$ and $eELSe'$ hold. Since $e^* < e'$, $\neg(e^*Re')$. This and e^*Re imply, by Axiom 4, that $\neg(eRe')$. Hence, there is e' such that eELSe' and $\neg(eRe')$ hold. ♥

(iii) $\forall e,e'(e \text{ R } e' \vee e' \text{ R } e)$.

Proof: Assume $\neg(eRe')$. Then, by Theorem 3(i), either $e < e'$ or $eELSe'$. If $e < e'$, then, by Axiom 2, $e'Re$. Suppose $eELSe'$ and $\neg(e'Re)$. The Theorem 1(i) implies $e'ELSe$. By Theorem 1(iii′), there is $e^{\#}$ such that $e^{\#} < e'$ and $e^{\#} < e$. $e^{\#} < e'$ implies, by Axiom 2, that $e'Re^{\#}$. $e'Re^{\#}$, the assumption $\neg e'Re$) and Axiom 5 imply that there is e'' such that $e^{\#} \leq e''$, $e'' < e$, $e'Re''$ and $e''Re'$. $e'Re''$ implies, by Theorem 2(iii), that either $e'' \leq e'$ or $e'ELSe''$. Since $e'' < e$ and $e'ELSe$, $e' \neq e''$. Since $e''Re'$, $\neg(e'' < e')$. Hence, $e'ELSe''$. By Axiom 2, $e'' < e$ implies eRe''. By Axiom 4, $e''Re'$ and eRe'' imply eRe'. This contradicts the assumption that $\neg(eRe')$. Hence, if $eELSe'$, then $e'Re$. ♥

Theorem 7.

(i) $\forall e,e'(e \text{ PRES } e' \rightarrow (e' = e \text{ or } e \text{ ELS } e'))$.

Proof: Assume $ePRESe'$. By Definition 4, eRe' and $e'Re$. By Theorem 2(iii), eRe' implies that either $e' \leq e$ or $eELSe'$. Assume $e' < e$. Then, Axiom 3 implies $\neg(e'Re)$. Contradiction. Hence, $e' = e$ or $eELSe'$. ♥

(v) $\forall e,e'((e \text{ PRES } e') \rightarrow (e \text{ R } e' \wedge \forall e''(e' < e'' \rightarrow \neg(e \text{ R } e''))))$.

Proof: $\Rightarrow$ Assume $ePRESe'$. By Definition 4 and Theorem 6(i), eRe' and $R_e = R_{e'}$. By Axiom 2, $\forall e''(e' < e'' \rightarrow \neg(e'Re''))$. Hence, $\forall e''(e' < e'' \rightarrow \neg(eRe''))$. $\Leftarrow$ Assume $eRe' \wedge \forall e''(e' < e'' \rightarrow \neg(eRe''))$. We need to show that $ePRESe'$. eRe' implies, by Theorem 2(iii), that either $e' \leq e$ or $eELSe'$. If $e' = e$, then Theorem 2(i) implies $e'Re$. This, together with the assumption eRe' and Definition 4, implies $ePRESe'$. If $e' < e$, then the second conjunct of

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the assumption implies $\neg$(eRe) which contradicts Theorem 2(i). Hence, eELSe′. By Theorem 6(vi), $\exists e^*(e' \leq e^* \wedge ePRESe^*)$. The assumption $\forall e''(e' \leq e'' \rightarrow \neg(eRe''))$ and Definition 4 imply $\neg(e' < e^*)$. Hence, $e' = e^*$. Since $e' = e^*$, ePRESe* implies ePRES e′.♥

Theorem 8.

(i) If $e' \in [e_1]$, $e'' \in [e_2]$ and $[e_1] \propto [e_2]$, then $\neg(e' \; R \; e'')$.

Proof: Assume $e' \in [e_1]$, $e'' \in [e_2]$, $[e_1] \propto [e_2]$ and $e'Re''$. By Definition 5, e_1PRESe' and e_2PRESe''. By Defintion 6, $\neg(e_1Re_2)$. e_2PRESe'' implies, by Definition 4, $e''Re_2$. This and the assumption $e'Re''$ imply, by Axiom 4, $e'Re_2$. e_1PRESe' implies e_1Re'. e_1Re' and $e'Re_2$ imply, by Axiom 4, e_1Re_2. Contradiction. Hence, $\neg(e'Re'')$.♥

(ii) For every e, set R_e is the union U of [e] and the causal pasts of all members of [e].

Proof: We first show that every element of R_e belongs to U. Assume $e^* \in R_e$, that is, eRe*. This, by Theorem 2(iii), implies that either $e^* \leq e$ or eELSe*. If $e^* = e$, then Theorem 7(ii) implies $e^* \in [e]$. Hence, $e^* \in U$. If $e^* < e$, e* is in the causal past of e. By Theorem 7(ii) and Definition 5, $e \in [e]$. Hence, $e^* \in U$. Finally, let eELSe*. Then, by Theorem 6(vi) and the assumption eRe*, $\exists e^{\#}(e^* \leq e^{\#} \wedge ePRESe^{\#})$. Thus, either e* is a member of [e] or e* belongs to the causal past of a member of [e]. Hence, $e^* \in U$.
For the other direction, suppose $e^* \in U$. If $e^* \in [e]$, then ePRESe* and eRe*. Hence, $e^* \in R_e$. Suppose $e^* < e'$ and $e' \in [e]$ for some e′. Then, ePRESe′ and eRe′. eRe′ and $e^* < e'$ imply, by Theorem 4(i), eRe*. Hence, $e^* \in R_e$.♥

(iii) $\propto$ is irreflexive.

Proof: By Theorem 2(i).♥

(iv) $\propto$ is transitive.

Proof: Assume $[e_1] \propto [e_2]$ and $[e_2] \propto [e_3]$. By Definition 5, that is equivalent to $\neg(e_1Re_2)$ and $\neg(e_2Re_3)$. $\neg(e_1Re_2)$ implies, by Theorem 5(iii), that e_2Re_1. By Axiom 4, e_2Re_1 and $\neg(e_2Re_3)$ imply $\neg(e_1Re_3)$. Hence, $[e_1] \propto [e_3]$.♥

(v) For every two equivalence classes $[e_1]$ and $[e_2]$, either $[e_1] \propto [e_2]$ or $[e_2] \propto [e_1]$ or $[e_1] = [e_2]$ holds.

Proof: Assume that $[e_1] \propto [e_2]$ and $[e_1] = [e_2]$ both do not hold. By Definition 5, the first assumption implies e_1Re_2. The second assumption implies $e_1 \neq e_2$. e_1Re_2, $e_1 \neq e_2$ and Theorem 2(iii) imply that either e_1ELSe_2 or $e_2 < e_1$. If $e_2 < e_1$, then Axiom 3 implies $\neg(e_2Re_1)$. Then, $[e_2] \propto [e_1]$. Assume that e_1ELSe_2 holds.

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Assume further that e_1PRESe_2. This implies that $[e_1] = [e_2]$. Contradiction. Hence, $\neg(e_1PRESe_2)$. This and e_1Re_2 imply, by Definition 4, $\neg(e_2Re_1)$. Hence, $[e_2] \propto [e_1]$.♥

(vi) $\propto$ is dense.

Proof: Assume $[e_1] \propto [e_2]$. We need to show that there is $[e_3]$ such that $[e_1] \propto [e_3]$ and $[e_3] \propto [e_2]$. The assumption and Definition 6 imply $\neg(e_1Re_2)$. This, by Theorem 3(i), implies that either $e_1 < e_2$ or e_1ELSe_2. Assume $e_1 < e_2$. Then, Axiom 1(vi) implies $\exists e_3(e_1 < e_3 \wedge e_3 < e_2)$. By Axiom 3, $\neg(e_1Re_3)$ and $\neg(e_3Re_2)$. Hence, $[e_1] \propto [e_3]$ and $[e_3] \propto [e_2]$. Assume e_1ELSe_2. By Theorem 1(i), e_2ELSe_1. $\neg(e_1Re_2)$ and Theorem 5(iii) imply e_2Re_1. e_2ELSe_1, e_2Re_1 and Theorem 6(v) imply $\exists e^*(e_2ELSe^* \wedge e_1 \leq * \wedge e_2PRESe^*)$. $\neg(e_1Re_2)$ and e_2PRESe^* imply $e_1 \neq e^*$. Hence, $e_1 < e^*$. Since e_2PRESe^*, $[e_2] = [e^*]$ and $e^* \in [e_2]$. Now, by Axiom 1(vi), $\exists e_3(e_1 < e_3 \wedge e_3 < e^*)$. By Axiom 3, $\neg(e_1Re_3)$ and $\neg(e_3Re^*)$. Hence, $[e_1] \propto [e_3]$ and $[e_3] \propto [e^*]$. Since $[e_2] = [e^*]$, it follows $[e_3] \propto [e_2]$.♥

(vii) $[e_1] \propto [e_2]$ iff $R_{e1} \subset R_{e2}$.

Proof: Assume $[e_1] \propto [e_2]$, that is, $\neg(e_1Re_2)$. By Theorem 4(iv), it follows that $R_{e1} \neq R_{e2}$. $\neg(e_1Re_2)$ implies, by Theorem 5(iii), that e_2Re_1. Assume now that $e \in R_{e1}$, that is, e_1Re. This and e_2Re_1 imply, by Axiom 4, that e_2Re. Hence, $e \in R_{e2}$ and $R_{e1} \subset R_{e2}$. For the other direction, suppose $R_{e1} \subset R_{e2}$, that is, $\forall e(e_1Re \rightarrow e_2Re)$ and $R_{e1} \neq R_{e2}$. By Theorem 2(i), $\forall e(e_1Re \rightarrow e_2Re)$ implies e_2Re_1. Theorem 4(iv), e_2Re_1 and $R_{e1} \neq R_{e2}$ imply $\neg(e_1Re_2)$. Hence, $[e_1] \propto [e_2]$.♥

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